

# Project Readme Template

Version 1 9/11/24

A single copy of this template should be filled out and submitted with each project submission, regardless of the number of students on the team. It should have the name `readme_teamname`

Also change the title of this template to "Project x Readme Team xxx"

| 1                | <b>Team Name:</b> Here in Theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                  |                       |            |  |             |                                                                                                                                                                                  |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |  |
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| 2                | <b>Team members names and netids:</b> Kaila Bryant (kbryant3) and Victoria Caruso (vcaruso)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |                       |            |  |             |                                                                                                                                                                                  |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |  |
| 3                | <b>Overall project attempted, with sub-projects:</b> SAT solving: brute force and backtracking                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |                       |            |  |             |                                                                                                                                                                                  |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |  |
| 4                | <b>Overall success of the project:</b> The project was successful!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                  |                       |            |  |             |                                                                                                                                                                                  |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |  |
| 5                | <b>Approximately total time (in hours) to complete:</b> 11-12 hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                       |            |  |             |                                                                                                                                                                                  |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |  |
| 6                | <b>Link to github repository:</b> <a href="https://github.com/victoriafcaruso/Project1-TOC">https://github.com/victoriafcaruso/Project1-TOC</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                  |                       |            |  |             |                                                                                                                                                                                  |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |  |
| 7                | <b>List of included files (if you have many files of a certain type, such as test files of different sizes, list just the folder): (Add more rows as necessary). Add more rows as necessary.</b> <table border="1"><thead><tr><th>File/folder Name</th><th>File Contents and Use</th></tr></thead><tbody><tr><td colspan="2">Code Files</td></tr><tr><td>team_sat.py</td><td>This file contains the code used to generate our output files which entailed reading in the inputs, parsing files, running the solvers, and creating our CSV and plotting files.</td></tr><tr><td>sat.py</td><td>This file contains our backtracking and brute force definitions (<b>backtrack</b>, <b>brute_force</b>, <b>sat_backtracking</b>, and <b>sat_bruteforce</b>), along with a definition that verifies if a given set of assignments for the variables could pass all clauses (<b>is_valid</b>). These functions are called on in the <i>run</i> definition in <code>sat_solver_helper.py</code> and the <i>run_solver_and_write_csv</i> definition in <code>team_sat.py</code>.</td></tr><tr><td colspan="2">Test Files</td></tr></tbody></table> | File/folder Name | File Contents and Use | Code Files |  | team_sat.py | This file contains the code used to generate our output files which entailed reading in the inputs, parsing files, running the solvers, and creating our CSV and plotting files. | sat.py | This file contains our backtracking and brute force definitions ( <b>backtrack</b> , <b>brute_force</b> , <b>sat_backtracking</b> , and <b>sat_bruteforce</b> ), along with a definition that verifies if a given set of assignments for the variables could pass all clauses ( <b>is_valid</b> ). These functions are called on in the <i>run</i> definition in <code>sat_solver_helper.py</code> and the <i>run_solver_and_write_csv</i> definition in <code>team_sat.py</code> . | Test Files |  |
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| team_sat.py      | This file contains the code used to generate our output files which entailed reading in the inputs, parsing files, running the solvers, and creating our CSV and plotting files.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                  |                       |            |  |             |                                                                                                                                                                                  |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |  |
| sat.py           | This file contains our backtracking and brute force definitions ( <b>backtrack</b> , <b>brute_force</b> , <b>sat_backtracking</b> , and <b>sat_bruteforce</b> ), along with a definition that verifies if a given set of assignments for the variables could pass all clauses ( <b>is_valid</b> ). These functions are called on in the <i>run</i> definition in <code>sat_solver_helper.py</code> and the <i>run_solver_and_write_csv</i> definition in <code>team_sat.py</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                       |            |  |             |                                                                                                                                                                                  |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |  |
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|   | team_tests.cnf                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | This file contains a dataset of SAT problems in CNF format in order to test and validate SAT solving methods, backtracking and brute force. The test inputs gradually increase in complexity as the number of clauses increase the further into the test inputs we go through.       |
|   | Output Files                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                      |
|   | backtracking_results.csv                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | This file contained the output results from our test cases, solving the test cases using backtracking and outputted the results in a CSV format with the column identifiers of "instance_id", "n_vars", "n_clauses", "method", "satisfiable", "time_seconds", and "solution"         |
|   | brute_force_results.csv                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | This file contained the output results from our test cases, solving the test cases using brute force and outputted the results in a CSV format with the column identifiers of "instance_id", "n_vars", "n_clauses", "method", "satisfiable", "time_seconds", and "solution"          |
|   | Plots (as needed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                      |
|   | plot_brute_vs_backtrack.png                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | This file contains an image of the plot that is generated based on the outputs from the inputs in our test file. The plot demonstrates the relationship between execution time and number of clauses and compares the results based on the brute force SAT solving and backtracking. |
| 8 | <b>Programming languages used, and associated libraries:</b> <ul style="list-style-type: none"> <li>- Used python with the following associated libraries: <ul style="list-style-type: none"> <li>- sys</li> <li>- os</li> <li>- csv</li> <li>- time</li> <li>- numpy as np</li> <li>- pandas as pd</li> <li>- matplotlib.pyplot</li> </ul> </li> <li>- sat.py also imports List, Tuple, and Dict from typing, and itertools (we did not use itertools and instead coded our functions in a way that was easier for us to break down and understand)</li> </ul> |                                                                                                                                                                                                                                                                                      |

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| 9  | <p><b>Key data structures (for each sub-project):</b></p> <ul style="list-style-type: none"> <li>- For functions in <a href="#">sat.py</a> (backtrack, brute_force, sat_backtracking, sat_bruteforce and is_valid) we used 2D lists to represent the <b>clauses</b> and <b>dictionaries</b> to represent <b>assignments</b>. <ul style="list-style-type: none"> <li>- Example of clauses: "[[-2, 3], [-1, 4], [-2, 4], [-4, -2], [-4, -1], [1, 1], [4, 4], [-4, -3], [-1, -4], [3, 3]]"</li> <li>- Example of Assignments: "{1: 0, 2: 0, 3: 1, 4: 1}" where the key is the variable, and each key points to either a 1 or 0 value of the assignment for that variable</li> </ul> </li> <li>- For functions in <b>team_sat.py</b>, we used tuples to represent single SAT problems from the input file (inst_id, n_vars, clauses). Additionally we used lists to keep track of results such as the number of variables (used for the x-axis in the plots) and times_list and sat_flags. Lastly, we also used csv writer objects to handle the formatting details of our CSV files.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 10 | <p><b>General operation of code (for each subproject)</b></p> <p>The function sat_backtracking calls on backtrack, and sat_bruteforce calls on brute_force, the latter of both are recursive (we chose to do recursion rather than use itertools given familiarity with recursion).</p> <p>The definition for <b>is_valid</b> is passed the list of clauses and an assignment for some number of variables. It goes through every clause in clauses, and checks if the current assignment passes each of them. An assignment can pass a clause if either a variable in the clause has not yet been assigned a value in assignment (i.e., the variable 4 appears in a clause but is not yet assigned 0 or 1 yet), OR, one of the assignments, say <math>x_1 = 0</math>, aligns with a variable in the clause (which usually has 2 variables but can have more). If a clause is not passed, False is returned, else, True is returned.</p> <p><b>sat_bruteforce</b> first defines a dictionary called <b>assignment</b> where every variable is represented by a key, like ints 1 or 2, and all variables are assigned to the value 0 (false). This initial assignment is passed to <b>brute_force</b> (with the initial depth 1 representing <math>x_1</math>), which goes through every possible <b>assignment</b> combination until one passes all the clauses with <b>is_valid</b>. It does this by trying assignments 0 and then 1 on the given variable (<math>x_1</math>), and for both, calling again on itself with the next depth (<math>x_2</math>) to try assignments 0 and 1 with both previous <math>x_1</math> values. For every assignment, the function checks if it is valid, and returns the assignment if so. The first valid assignment combination will be returned by the call stack, and if the problem is unsolvable, {} is eventually passed back.</p> <p><b>sat_backtracking</b> defines an empty dictionary of assignments, and passes it along with depth = 1 to backtrack. backtrack is similar to the above, except it uses pruning to see if adding an assignment, like <math>x_2 = 0</math>, keeps the dictionary of assignments valid. For example, say <math>x_1 = 0</math> in the passed assignment ({1:0}), if adding <math>x_2 = 0</math> ({1:0, 2:0}) causes the <b>assignment</b> to then fail <b>is_valid</b>, then the function tries adding <math>x_2 = 1</math> ({1:0, 2:1}). If this passes, the function moves onto assigning a value to <math>x_3</math> by calling on itself with a depth = 3 and the new assignment, else, it back tracks (in the call stack) and tries <math>x_1 = 1</math>. Once an assignment with an n_vars number of keys passes <b>is_valid</b>, it is returned.</p> <p><b>run_solver_and_write_csv</b> focuses on the following operations; creating an output directory (if one does not already exist), initializes empty lists to keep track of all</p> |

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|    | <p>necessary data (number of variables, execution times, and satisfiability flags, opens a CSV file for writing and writes to that CSV file for each instance.</p> <p><b>read_team_inputs</b> again, creates a results folder if it does not already exist, calls the helper function “<code>parse_multi_instance_dimacs()</code>” in order to parse the input files into problem instances. It then creates a <code>SatSolver</code> object that is initialized with the input file and determines output paths for both CSV files (one for brute force and one for backtracking results). Next, it runs both solving methods on all problem instances and writes the results to the appropriate CSV file. The function will then return the paths for both CSV files.</p> <p><b>plot_brute_vs_backtrack</b> and <b>plot_method</b> were both used in our plotting efforts. <b>plot_brute_vs_backtrack</b> reads both CSV files and plots the results of the satisfiability of the test inputs and saves the plot to a PNG file. <b>plot_brute_vs_backtrack</b> also uses <b>plot_method</b>, a helper function, to plot the data points and the best-fit curve for each solving method</p> |
| 11 | <p><b>What test cases you used/added, why you used them, what did they tell you about the correctness of your code.</b></p> <ol style="list-style-type: none"> <li>1. <code>team_tests.cnf</code> <ol style="list-style-type: none"> <li>a. We used about 1000 lines from the <code>KSAT.cnf</code> file from Canvas to create our test input file. We used this input in order to test our <code>team_sat.py</code> file and have more data points to graph in the resulting <code>plot_brute_vs_backtrack.png</code> graph. These tests showed that our code for both methods (backtracking and brute force) were able to correctly identify solvable and unsolvable problems, and the backtracking method was much quicker than the brute force method, especially as the number of variables in a problem increased.</li> </ol> </li> </ol>                                                                                                                                                                                                                                                                                                                                              |
| 12 | <p><b>How you managed the code development</b></p> <p>We managed code development by meeting in person and discussing our division of work. We divided the work and checked in with each other to ensure neither was stuck or struggling and talked out issues with each other. One of us focused on implementing the backtracking and brute force algorithm, while the other worked on implementing the plotting algorithm and creation/reading of our CSV files.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 13 | <p><b>Detailed discussion of results:</b></p> <p>We were able to successfully design both brute force and backtracking functions that could correctly identify when problems were unsolvable and, if problems were solvable, assignments to the problem variables that solved a given SAT problem. We were able to plot this information to display how quick each program ran (time) as the number of variables in the problem increased; the graph also distinguishes when a problem was solvable or unsolvable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 14 | <p><b>How team was organized</b></p> <p>While we collaborated on every part of the assignment, Victoria was mainly responsible for coding the backtracking and brute force definitions in <a href="#">sat.py</a> (after we wrote the pseudo code for this together) and Kaila was mainly responsible for <code>team_sat.py</code> (after we defined the requirements and necessary definitions for this file). For most of the time when one person was coding on their laptop, we were working together and discussing</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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|    | ways to write any given portion of code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 15 | <p><b>What you might do differently if you did the project again</b></p> <p>If we did the project differently, we could have used <b>itertools</b> rather than <b>recursion</b> to write our <b>bruteforce function(s)</b>. This would have reduced the number of lines of code we had to solve the SAT problems with the bruteforce method; we could have looped through “<code>itertools.product([0, 1], repeat=n_vars):</code>” to generate all possible assignments rather than recursively calling on <code>brute_force</code> to assign a value to each variable.</p> |
| 16 | Any additional material:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |